

Time Series with Konrad: episode 1

The basic machinery: trend, seasonality, and the rest



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Welcome to **Time Series with Konrad**, a series on practical time series methods.

In this first post, we'll cover the essentials: what time series data is, how to break it into meaningful pieces, and introduce a few core ideas - concepts like dependence and stationarity matter long before you choose a forecasting model.



All code lives in the companion notebook:

<https://github.com/tng-konrad/time-series-with-Konrad/blob/main/m01-the-basics.ipynb>

Groundwork

A *time series* is any sequence of data points recorded over time. That's it, that's a formal definition. *thanks for coming to my TED talk ;-)*

Time series are everywhere: tracking crypto prices, monitoring weather patterns, analyzing website traffic, recording the rhythm of your heartbeat – if something changes as time passes and we measure it, we have a time series.

More formally, time series data consists of observations taken at specific points in time. These intervals between observations are often regular: hourly temperatures, daily sales figures, weekly unemployment

rates, monthly stock returns - you get the idea. Irregular time intervals are possible as well, though they require slightly different handling.

DISCLAIMER: I promise to keep the jargon to the necessary minimum throughout this series - but a little bit is necessary.

We will denote a basic time series as:

$$\{X_t\} \quad t = 1, \dots, T$$

Here T denotes the total number of observations (*at some point we stop recording and start thinking what to do with this stuff*), while t is an index counting each measurement in order.

Formally, time series are studied using probability theory, but we won't need that machinery here.

So, what can we **do** with time series data? Why bother analyzing it? Here are some common applications:

- **Interpretation:** Understanding the underlying dynamics driving observed phenomena. We want to make sense of complex patterns and identify key trends.
- **Modelling:** Developing a representation of the time series that captures its inherent characteristics, allowing us to understand **how** the data behaves.
- **Forecasting / Prediction:** Estimating future values based on past observations – essentially, looking into the future. *The two terms can be used interchangeably and pretty much reflect the background of the author (engineering / econometrics).*
- **Filtering / Smoothing:** Reducing noise and revealing underlying patterns in the data, providing a clearer picture of the process being observed. *This is useful when dealing with incomplete or noisy measurements.*

- **Simulation:** Generating artificial time series to explore potential scenarios or estimate probabilities that are difficult to calculate directly from limited historical data (e.g. estimating extreme risk events).

Decomposition

A useful first step is to break a time series into simpler components with clear interpretation. This is called **decomposition** and it helps us isolate different patterns and gain insights into the underlying dynamics. A simple representation is::

$$X_t = T_t + S_t + I_t$$

where:

- The trend **T** represents the long-term direction of the time series. It shows whether the data is generally increasing, decreasing, or remaining stable over an extended period. *Think of it as a smooth progression underlying the short-term fluctuations.*
- The seasonal component **S** captures repeating patterns that occur at fixed intervals. These patterns are often linked to calendar effects: increased retail sales during the holiday season, higher electricity demand in summer months, or weekly variations in website traffic. The period (length of the interval) is **known and consistent**.
- The residuals a.k.a. irregular component **I** represents the random fluctuations or noise remaining after removing the trend and seasonal components. *It captures variations that aren't explained by these systematic patterns.*

The relationship between these components can vary: we often model time series as either *additive* or *multiplicative*, depending on how the components combine. Additive decomposition works well when

seasonal effects are roughly constant over time. When seasonal fluctuations grow as the series grows, a **multiplicative** structure is often more appropriate:

$$X_t = T_t * S_t * I_t.$$

You can obviously mix and match the relationship, e.g. have an additive one between trend and seasonality, with residuals proportional to the level:

$$X_t = (T_t + S_t) * I_t$$

In practice, decomposition estimates the trend using smoothing, extracts seasonality by averaging recurring patterns, and treats what remains as noise.

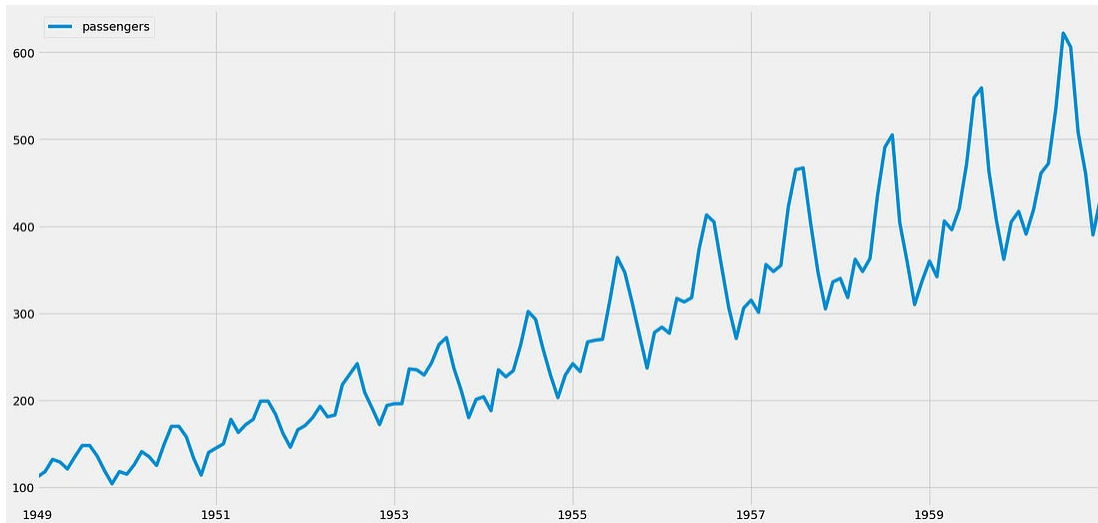
Let's try some examples.

Passengers dataset

Every field of knowledge has **the** dataset that is used for teaching purposes: machine learning has Iris and CIFAR, differential equations - Canadian lynx data, and statistics has the airline passengers dataset between 1949 and 1960, first compiled by Box and Jenkins (*you will be hearing those two names again in the next module*) in 1976.

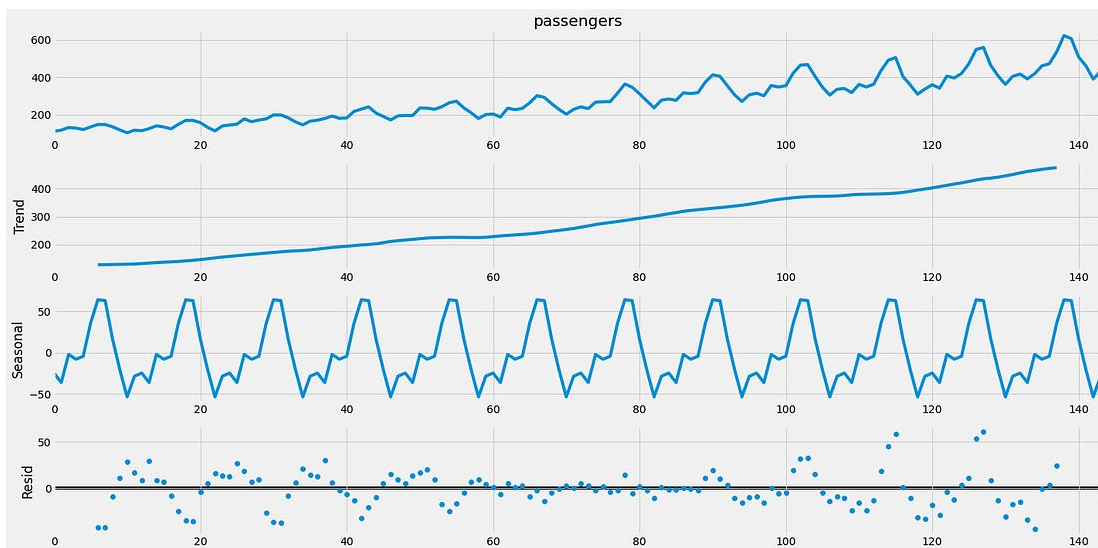
We will use this dataset to demonstrate in practice what kind of information can be obtained using seasonal decomposition.

PSA: in the main text of the tutorial, I am only using chunks of the code needed for demonstration. For a complete code explanations, please consult the accompanying notebook linked at the beginning.



Let's start with a basic additive decomposition:

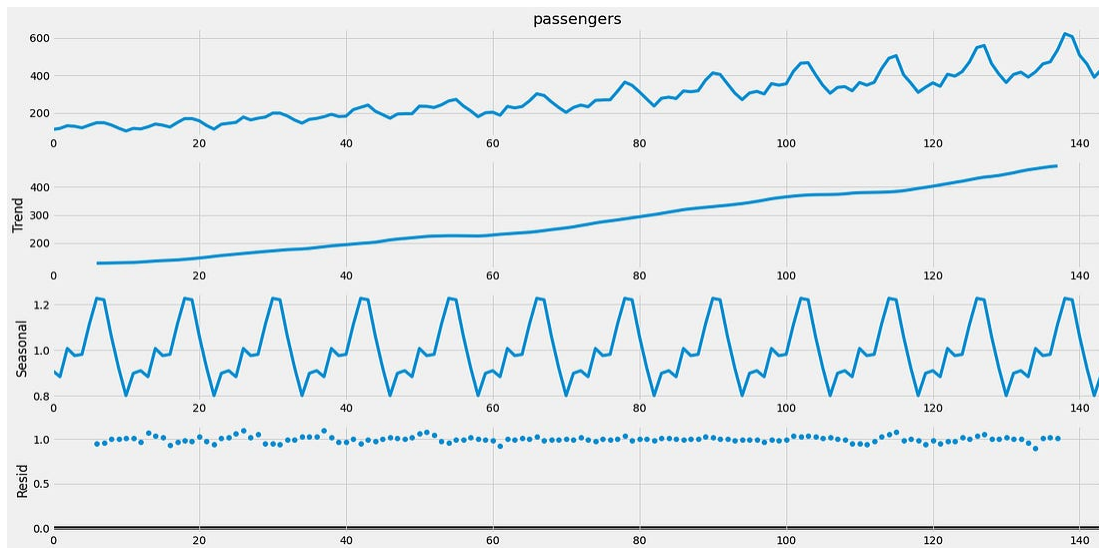
```
decomposition1 = seasonal_decompose(series["passengers"], period = 12)
```



Trend and seasonality are behaving more or less in line with expectations, but the behavior of the residuals is clearly not consistent over time (average level of oscillations in the middle of the sample is very different than on either end). While there are many possible reasons, one quick explanation is the additive vs multiplicative

relationship between the series components - which is something we can examine quickly:

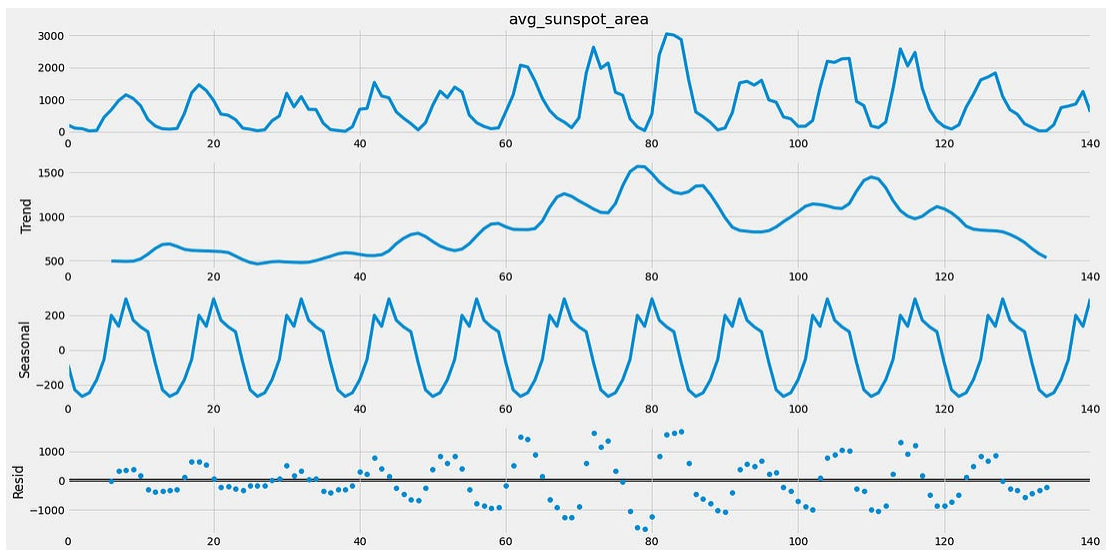
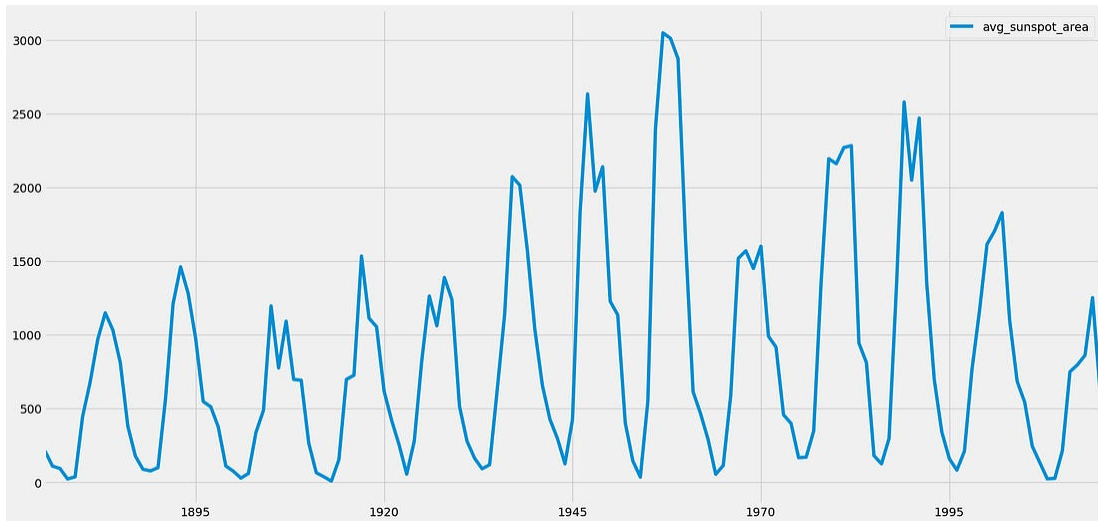
```
decomposition2 = seasonal_decompose(series["passengers"], period =12, model =  
'multiplicative')
```



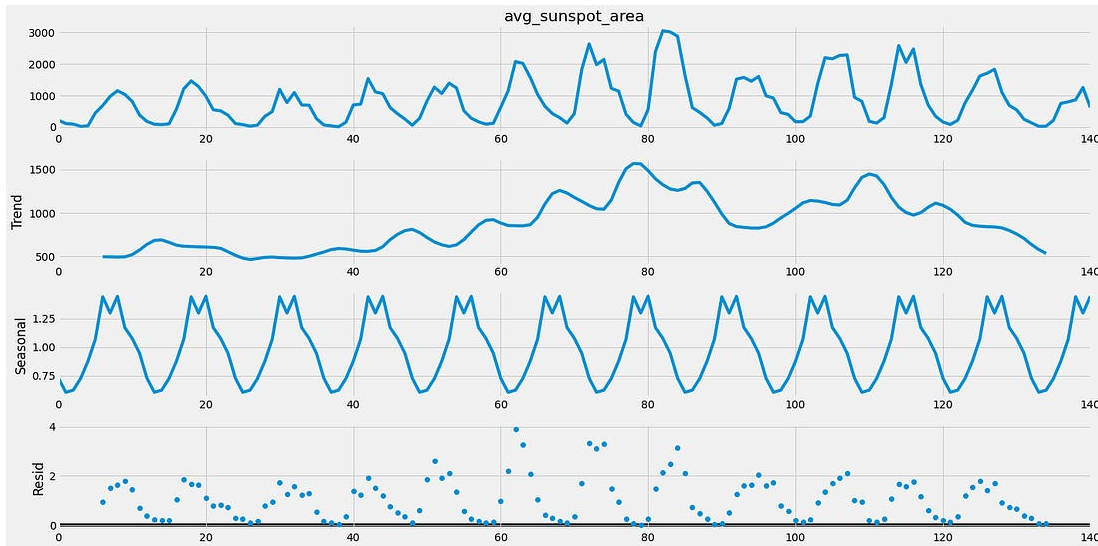
Not much of a qualitative change in trend and seasonality components, but the residuals looks much more stable around a constant level.

Sunspots dataset

Not all data behaves this cleanly. Sunspot activity shows cycles, but their timing and strength vary.



Additive decomposition is clearly not sufficient to explain what's going on in the data. How about multiplicative?



As we can see, changing the decomposition type alleviates the problem with residual behavior, *but only to a certain degree* - the pattern is still not consistent, with increase in the amplitude in the middle of the sample. This demonstrates that while seasonal decomposition is a fast tool, it has severe limitations when dealing with more sophisticated data generating processes.

Dependence

The only way to successful prediction is if past values of a series carry some information about the future behavior - in other words, if the present values are dependent on the past. A fast - and therefore very popular - manner of examining this dependence are the autocorrelation and partial autocorrelation functions.

Autocorrelation: overall correlation with past values.

$$\rho(u, t + u) = \text{Cor}(X_u, X_{t+u}) = \frac{\text{Cov}(X_t, X_{t+u})}{\text{Var}(X_t)\text{Var}(X_{t+u})}$$

For pure random noise, autocorrelations should be close to zero at all lags.

When this is not the case:

- Slowly decaying correlations often indicate trends.
- Repeating peaks suggest seasonality.
- Sharp spikes at specific lags indicate structured dependence.

Partial autocorrelation: direct correlation after removing intermediate effects.

$$\phi(u) = \text{Cor}(X_t, X_{t+u} | X_{t+1}, \dots, X_{t+u-1})$$

The PACF isolates *direct* effects. Significant spikes often point to how many past observations matter for prediction.

Quick takeaway:

- **ACF** includes both direct and indirect effects.
- **PACF** removes intermediate effects and shows only the direct link.

Stationarity



The reason to bother with the concept in stationarity can be best summarized with a bad joke I heard many years ago as a math student: say, a mathematician is given a piece of wood with two nails - one hammered all the way in, the other half there - and told to get remove them. How does he solve it?

1. start with the nail that's completely in - that's the more interesting case
2. once done, hammer the other all the way - to reduce to an already solved case

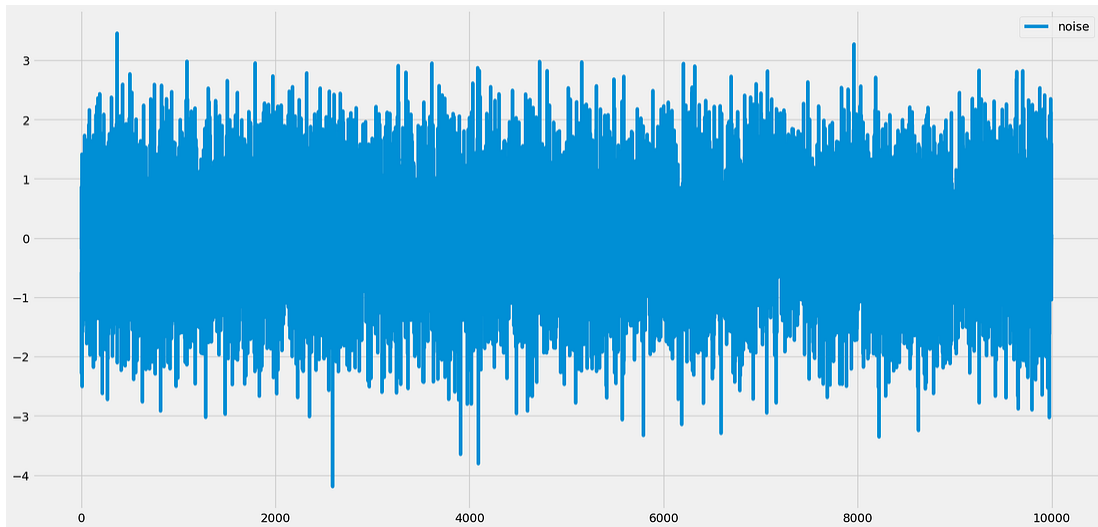


I warned you it was bad.

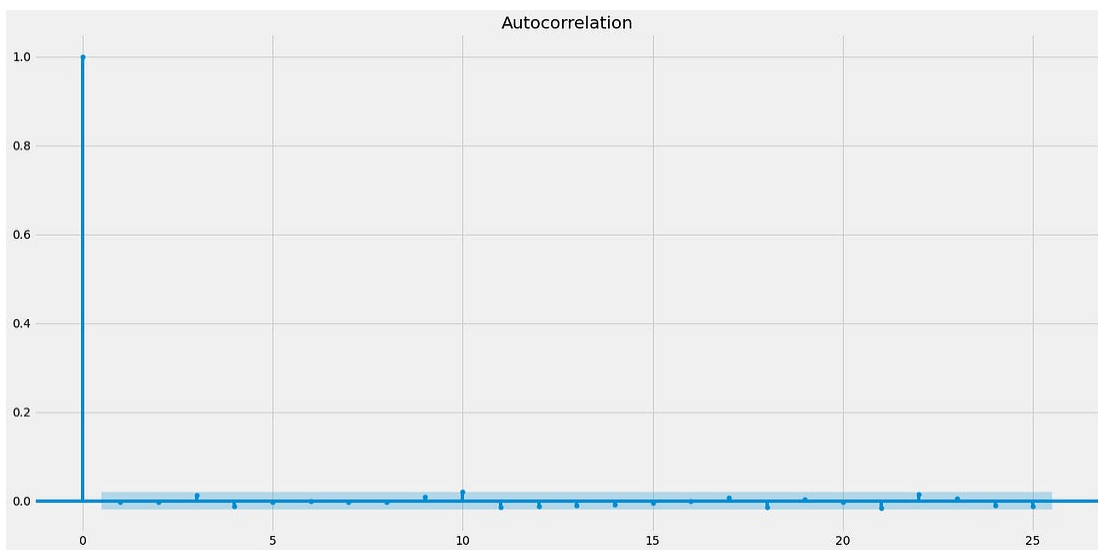
Informally, a series is stationary if its average level, variability, and dependence structure remain stable over time. Models trained on stable data behave more predictably and are easier to validate and explain.

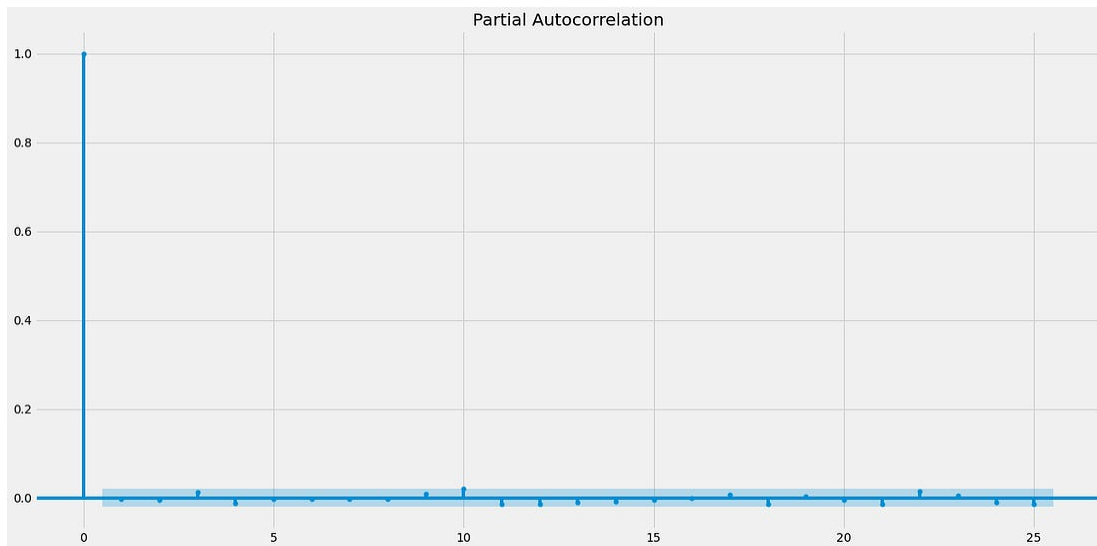
A useful way to think about this is that we often try to transform messy data into a simpler, more stable form before modeling it.

What is the simplest non-trivial stationary series? Gaussian white noise:



What do ACF / PACF tell us?





As expected, we get values close to zero at all lags. Random noise has no memory (= *today tells us nothing about tomorrow*), so correlations should look like random scatter around zero.

For a slightly more interesting example (of non-stationary behavior), we can examine the passengers dataset (*which we won't plot again to save space ;-)*

The stationarity of a series can be checked by examining the distribution of the series: we split the series into 2 contiguous parts and compute the summary statistics like the mean and, variance.

mean:

chunk1: 182.90 vs chunk2: 377.69

variance:

chunk1: 2244.09 vs chunk2: 7367.96

The values are clearly very different across the two data subsets, which strongly suggests non-stationarity.

(A little bit of) theory

A process is (weakly) stationary if it satisfies the following conditions:

- constant mean
- covariance only depends on distance in time between variables
- finite variance

So how can we turn a non-stationary series into a stationary one?

Popular transformations include (but are not limited to):

- differencing the series
- taking the log of the series
- power transforms

Differencing

Differencing can help stabilize the mean of a time series by removing changes in the level of a time series, and so eliminating trend and seasonality.

It's important to distinguish between **differencing multiple times** and **differencing once at a larger lag**, because they solve different problems. Applying first differences repeatedly is mainly used to remove *trends*. Each application removes one degree of polynomial trend: a linear trend disappears after one difference, a quadratic trend after two, and so on.

In contrast, differencing once at lag n compares each observation to its value n periods earlier. This does not target long-term trends, but removes **seasonal patterns** that repeat every n steps (for example, annual seasonality in monthly data with $n = 12$).

Although both operations involve subtracting past values, they have very different effects: **repeated differencing simplifies long-run growth, while seasonal differencing cancels repeating cycles.**

Tests for stationarity

While inspecting plots before / after transformation can be useful to assess presence of trends or seasonalities (as we did above with the passengers dataset), in practice we need a more formal approach - like testing a hypothesis (*introduction to statistical tests is beyond the scope of this notebook, so if you feel like you need a refresher, please consult other sources*). The most popular tests dealing with stationarity are:

- Augmented Dickey-Fuller (ADF)
- Kwiatkowski-Phillips-Schmidt-Shin (KPSS)
- Philips-Perron (PP)

ADF test is a unit root test: it determines how strongly a time series is defined by a trend. It helps answer a practical question: *Do we need to transform this series before modeling it?*

- Null hypothesis (H0): the time series can be represented by a unit root that is not stationary.
- Alternative hypothesis (H1): the time series is stationary.

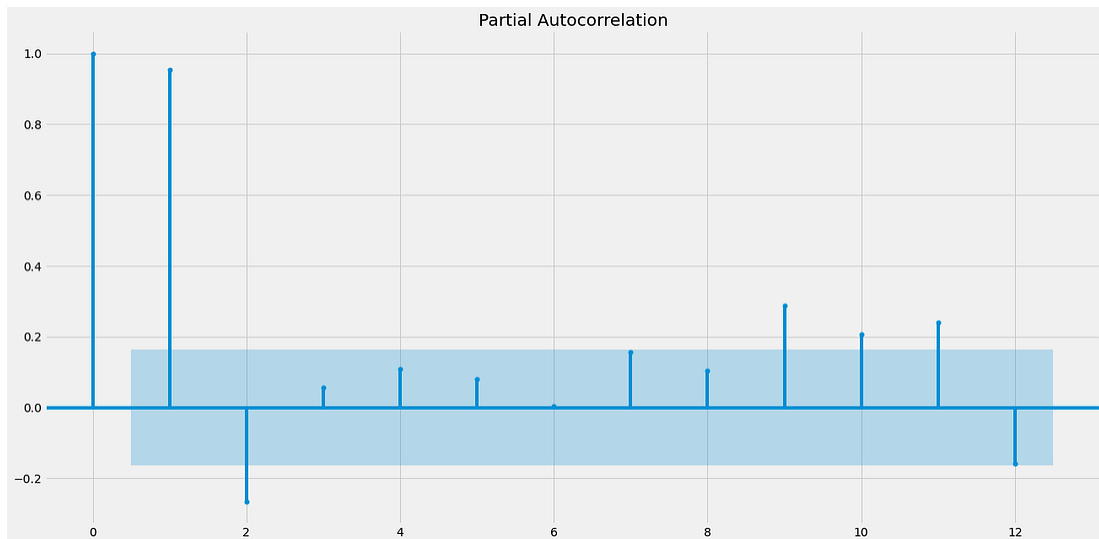
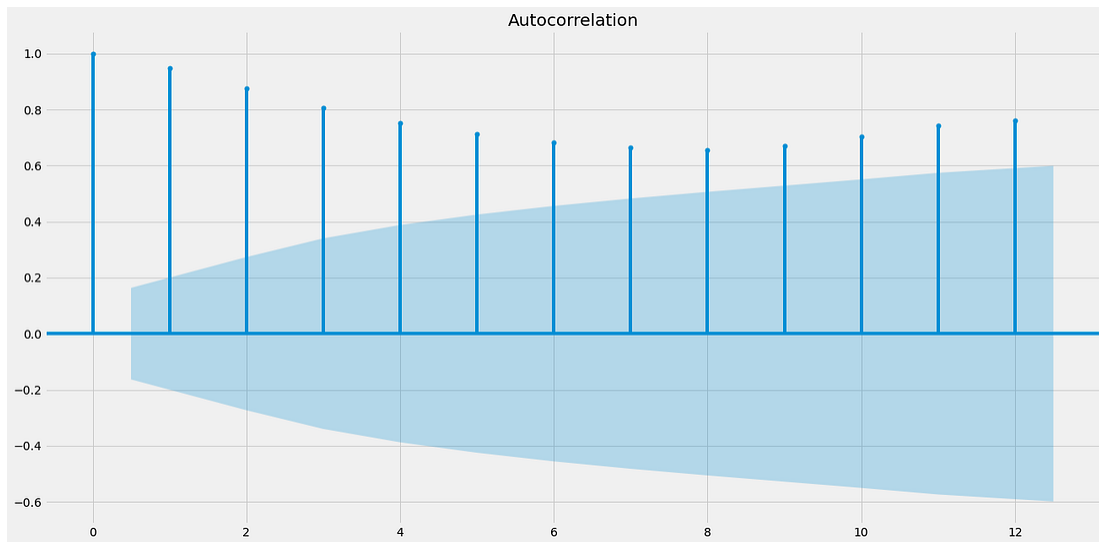
Interpretation of p-value: how compatible the data is with the null hypothesis:

- $p < 0.05 \Rightarrow$ reject non-stationarity, the data is stationary
- $p \geq 0.05 \Rightarrow$ the data has a unit root, non-stationarity remains plausible

In short,

An important caveat: remember that statistical tests do not accept a hypothesis - we can only fail to reject it.

Let's have a closer look at the dependence structure in the passenger dataset:



ACF / PACF suggest it is not stationary, and the ADF results confirm it:

ADF Statistic: 0.815369

p-value: 0.991880

We can decompose the series to check the components one by one:
which parts are responsible for the non-stationary behavior?

Trend:

p-value: 0.995

Seasonal:

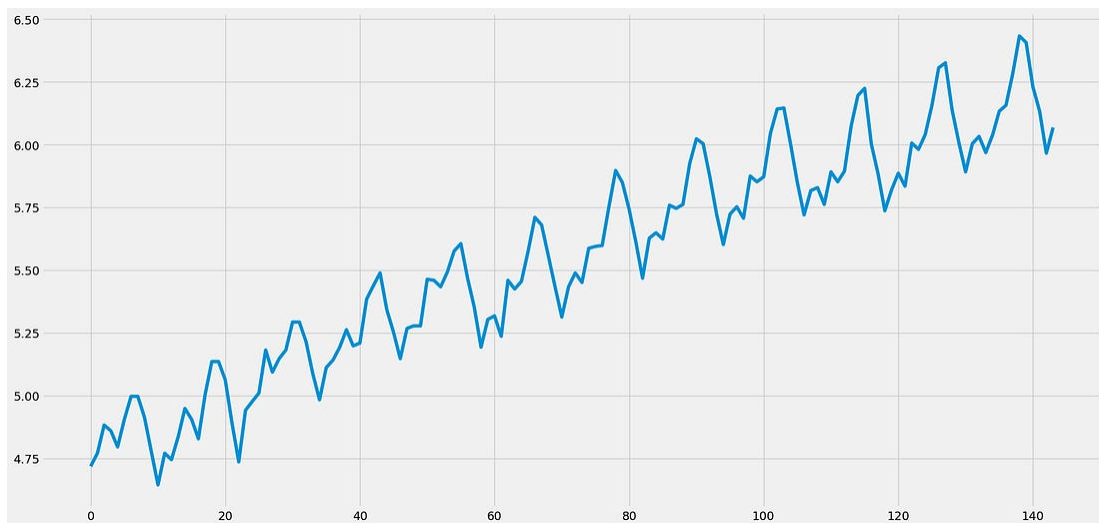
p-value: 0.000

Residuals:

p-value: 0.000

As we can see, the hypothesis of non-stationarity is rejected for the trend component, but not for seasonal. The reason for that is that ADF test check for a very specific form of non-stationarity, namely variation in the presence of a linear trend - while the seasonal component is clearly not stationary (see graph above), it is a qualitatively different kind of behavior.

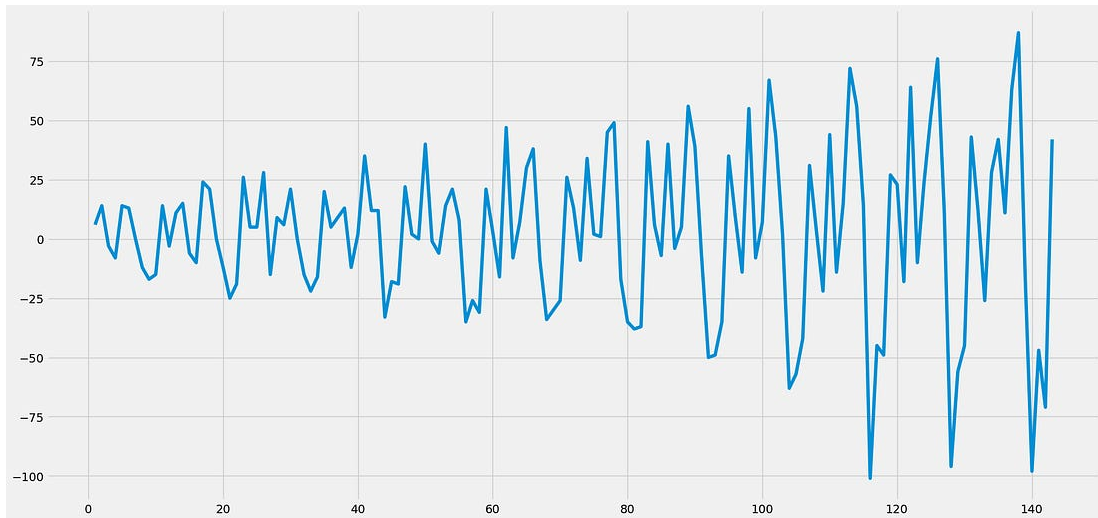
Let's try some transformations to make the series stationary: first, apply a logarithm.



The transformation does not remove the trend, but it does seem to stabilize the amplitude (periodic variations have comparable magnitude now). How does that translate into ADF results?

p-value: 0.422

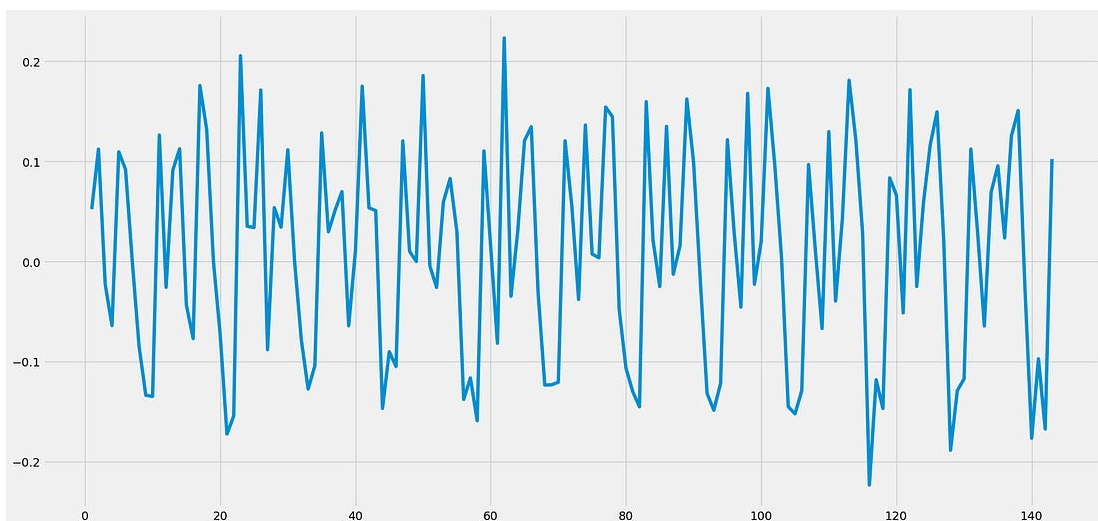
The null is still not rejected, but p-value has dropped - which indicates the transformations are the right way to go. Next, we can try differentiating to get rid of the trend



As expected, differentiation removes the trend (oscillations happen around a fixed level), but variations amplitude is magnified.

p-value: 0.054

We continue moving in the right direction - what happens if we combine the two transformations?



p-value: 0.028

So after applying logarithm (to stabilize the variance) and differentiation (to remove the trend), we have transformed our series to one that can be plausibly treated as stationary.

Closing time

Before choosing sophisticated models, it pays to understand the structure of your data. Trends, seasonality, dependence structure - they determine what methods will work and which ones will fail. Getting these basics right often matters more than model complexity.

In the next post, we'll build on this foundation and start turning these ideas into concrete forecasting models. Stay tuned.

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